

InsightSwarm — A Complete Plain Language Guide

Everything you need to know, explained simply

Bharat College of Engineering, University of Mumbai, 2026

This guide is written for **everyone** — students, family members, professors, curious people — not just programmers. No technical background needed.

Read it top to bottom, or jump to any section that interests you.

Section 1 — Why Did We Make This Project?

Think about the last time you saw a "fact" shared on WhatsApp or Facebook. Maybe it said *"Drinking hot water cures cancer"* or *"The government is putting chips in vaccines."* These messages spread to millions within hours. Most people forward them without checking.

The problem is huge:

- Over **10,000 pieces** of misinformation are shared every hour on social media.
- A lie travels **six times faster** than the truth online (proven by MIT research).
- Real fact-checking organisations like Snopes or AFP Fact Check employ human journalists — a team of 20 can only check **10–15 claims per day**. That is nowhere near enough.

So what about AI? Can't AI just check facts automatically?

The short answer is **yes**, but existing AI tools have a serious problem called **"hallucination."** This means the AI sometimes makes up fake sources that sound real. It might tell you *"According to a 2023 study in the Journal of Medicine, coffee cures diabetes"* — and that study simply does not exist. **About 20–23% of AI fact-checking responses contain made-up sources.**

We made **InsightSwarm** to solve **both** problems at once:

1. Automate fact-checking so it can handle many claims quickly.
2. **Eliminate fake sources** by actually visiting every single link the AI uses as evidence.

We also wanted it to be **completely free** to use — no subscription, no server bills. It runs entirely on free tools and free API tiers, forever.

Section 2 — What Is InsightSwarm?

InsightSwarm is an AI-powered fact-checking system. You type in a claim — any statement you want to verify — and the system tells you whether it is:

- ☐ TRUE
- ☒ FALSE
- ☐ PARTIALLY TRUE
- ☐ INSUFFICIENT EVIDENCE

But here's what makes it different from simply asking ChatGPT:

*InsightSwarm does **not** rely on just one AI. Instead, it sets up a structured **debate** between multiple AI "agents" (think of them as AI workers, each with a specific job). These agents argue, fact-check, and then a **judge agent** looks at everything and gives a final verdict.*

The four main agents:

1. The Supporter (ProAgent)

*Finds the best possible arguments **FOR** the claim being true.*

2. The Challenger (ConAgent)

*Finds the best possible arguments **AGAINST** the claim.*

3. The Fact Checker (FactChecker)

*Takes every single web link mentioned and actually **visits** those websites to verify content. This is the key anti-hallucination layer.*

4. The Judge (Moderator)

After seeing all arguments and verified sources, weighs everything and announces the final verdict with a confidence score.

The whole debate runs automatically in about **35–60 seconds**.

Simple example:

You type: "Coffee prevents cancer"

- **Supporter** argues why this might be true (links coffee to reduced colorectal cancer risk in some studies).
 - **Challenger** argues why this might be false (the claim is too broad, evidence is limited).
 - **Fact Checker** visits every website mentioned; some links are real, some are made up.
 - **Judge** looks at everything and might say:
 - ☐ **PARTIALLY TRUE** — "There is some evidence linking coffee to reduced risk of certain cancers, but the claim overstates the science. 68% confidence."
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Section 3 — How Does It Work? (The Full Journey of a Claim)

Here's exactly what happens from the moment you press **Enter** to when you see the result.

Step 1 – You Submit a Claim

You open the web interface (built in React) and type something like:

"The Earth is getting warmer because of human activity."

Step 2 – Have We Seen This Before? (Smart Cache)

Before doing any work, InsightSwarm checks its memory (an SQLite database). If someone already asked a **very similar question** recently, it returns the answer immediately — in **under 1 second**.

It determines similarity by converting your question into numbers (embeddings) and comparing them mathematically. If similarity is $\geq 85\%$, it reuses the cached answer.

Step 3 – How Complicated Is This Claim?

A special module, the **Claim Complexity Estimator**, scores how hard the claim will be to verify. It checks:

- Technical topic? (medicine/law → harder)
- Predictions about the future? (harder)
- Many named entities? (people, places, organisations → harder)
- Simple everyday fact? (easier)

Based on the score, the system decides how many debate rounds to run (2–4 rounds).

Step 4 – Getting Real Evidence First

Before the debate starts, the system searches the web using the **Tavily API**. It gathers real web articles — some supporting, some opposing — and hands them to the debating agents so they have actual evidence.

Step 5 – The Debate

Round by round, the Supporter and Challenger take turns making arguments. Each agent must **respond directly** to what the other said in the previous round. Every argument comes with links to supporting sources.

Step 6 – Verifying Every Source

This is the most important step. The **Fact Checker** agent takes every single link and:

Check	Action
a) URL exists?	Tries to open the website. If it fails → flagged as hallucinated source .
b) Content matches?	Reads the page content and mathematically compares it to the claim. If the content does not support what was claimed → flagged as mismatch .
c) Additional checks	Paywall? Trustworthy domain? Date valid?
d) Trust score	Assigns a score based on the source type.

Example trust scores:

- Government websites (.gov): **0.85**
- Academic journals, NASA, WHO: **0.90**
- Reuters, BBC, AP News: **0.75**
- Wikipedia, general websites: **0.50**
- Social media, blogs: **0.30**
- Known misinformation sites: **0.10**

Step 7 – Human Check (If Needed)

If the verification rate is very low (too many sources failed), the system pauses and shows a **Human-in-the-Loop** screen. A human can review each source and override the AI's assessment before the final verdict.

Step 8 – The Final Verdict

The Judge calculates a final confidence score using this formula:

$$\begin{aligned}
 \text{Confidence} = & \\
 & (\text{Quality of Arguments} \times 30\%) + \\
 & (\text{How many sources were verified} \times 30\%) + \\
 & (\text{Average trustworthiness of sources} \times 20\%) + \\
 & (\text{Scientific consensus score} \times 20\%)
 \end{aligned}$$

Step 9 – You See the Result

The website shows:

- Final verdict in a coloured box (green=TRUE, red=FALSE, yellow=PARTIAL)
 - Confidence percentage
 - Full debate transcript
 - Table of every source checked (pass/fail status)
 - Explanation of why the verdict was reached
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Section 4 — The Architecture, in Simple Words

"Architecture" just means how the system is built. Think of it like the floor plan of a building.

InsightSwarm has four main floors:

Floor	Name	Description
1	What You See (The Website)	React interface with three panels: claim input, live debate, and results.
2	The Front Door (API Server)	FastAPI server receives requests, validates them, and enforces rate limits.
3	The Brain (Debate Engine)	Built with LangGraph; manages the debate workflow and the four AI agents.
4	The Foundation (AI Providers & Storage)	Talks to Groq, Google Gemini, Cerebras, OpenRouter. Stores data in SQLite.

Section 5 — The Technology and Software Used

AI Thinking

- **Groq API** – Runs Meta’s Llama 3.3 70B (very fast)
- **Google Gemini 2.5 Flash** – Opposing agent and judge
- **Cerebras / OpenRouter** – Backup providers
- **LangGraph** – Workflow orchestration

Website (Frontend)

- **React 18 + Vite**
- **Zustand** – State management

Backend (Server)

- **FastAPI** (Python 3.11)
- **Pydantic** – Data validation

Storage & Caching

- **SQLite**
- **sentence-transformers** – Embeddings for smart cache

Source Checking

- **BeautifulSoup4** – Web scraping
- **RapidFuzz** – Text similarity
- **Tavily API** – Web search
- **scikit-learn** – Cosine similarity for content matching

Reliability

- **tenacity** – Automatic retries
- **slowapi** – Rate limiting

Total Code Size: Over 15,600 lines
Tests Written: 168+ automated tests (all passing)
Cost to Run: Rs. 0 (free tiers only)

Section 6 — Why Is InsightSwarm Better Than Existing Systems?

Comparison Table

Metric	Keyword System	Single AI	InsightSwarm
Accuracy (F1 Score)	0.56	0.68	0.81
Precision	0.54	0.70	0.82
Hallucination Rate	N/A	~20%	< 3%
Calibration Error (ECE)	N/A	0.31	0.24
Cost per month	Rs. 0	Rs. 0	Rs. 0

Five Original Research Contributions

1. **Human-in-the-Loop via AI Graph Interrupts** – The debate graph can pause and wait for a human reviewer.
2. **Adaptive Confidence Calibration** – Corrects the AI’s natural under-confidence bias.
3. **Argumentation Quality Analysis** – Detects **10 types of logical fallacies** in the agents’ arguments.
4. **Claim Complexity Estimation** – Adjusts the number of debate rounds based on difficulty.
5. **Explainability Engine** – Shows exactly which factors influenced the confidence score.

Section 7 — Common Questions and Answers

Q: Do I need to pay for this?

A: No. It runs entirely on free API tiers. Groq gives 14,400 free requests/day; Gemini gives 250/day; Tavily gives 1,000 free searches/month.

Q: How fast is it?

A: New claim → 35–60 seconds. Cached claim → < **1 second**.

Q: How accurate is it?

A: In testing on 100 fact-checking tasks, **81%** matched professional fact-checkers' decisions.

Q: What if a website is down?

A: Each source has a 10-second timeout. Failed sources are marked as **TIMEOUT** and do not affect the verdict.

Q: Can it be fooled?

A: Tested against adversarial inputs. It returns **INSUFFICIENT EVIDENCE** rather than giving a confidently wrong answer.

Q: What languages does it support?

A: Currently English; multilingual support is planned.

Q: What does "PARTIALLY TRUE" mean?

A: The claim is not completely right or completely wrong. Example: "*Coffee prevents cancer*" — some evidence exists but the claim is overstated.

Q: What is a "hallucinated source"?

A: When AI invents a URL that doesn't exist (Type I) or a real URL whose content doesn't match what the AI claimed (Type II). InsightSwarm catches both.

Q: How is it different from Google Fact Check?

A: Google Fact Check only shows *existing* human fact-checks. InsightSwarm creates a brand-new fact-check from scratch.

Q: Who built this?

A: Students at Bharat College of Engineering, University of Mumbai:

- Soham Gawas
- Bhargav Ghawali
- Mahesh Gawali
- Ayush Devadiga

Under the guidance of **Prof. Shital Gujar**, Dept. of CSE (AI & ML). Published in IJRASET (2026).

Q: Is InsightSwarm open source?

A: Yes — github.com/AyushDevadiga1/Insight-Swarm

Section 8 — A Simple Example from Start to Finish

Claim: *"Vaccines cause autism"*

1. **Cache check** – Possibly seen before; if new, proceeds.
2. **Complexity** – Medical topic → marked as **HIGH** complexity (3 rounds).
3. **Consensus check** – Overwhelming scientific consensus against the claim. Flagged as **SETTLED SCIENCE**; confidence set to **99% FALSE** immediately.
4. **Supporter** argues: cites the debunked 1998 Wakefield study.
5. **Challenger** argues: Wakefield study retracted, fraud, loss of medical license, over 1 million children studied with no link.
6. **Fact Checker** verifies: retraction notice verified on Lancet.com; CDC statements verified.
7. **Judge** calculates: 99% confidence FALSE.

Result Displayed:

☐ **FALSE** (99% confidence)

"This claim has been thoroughly debunked by decades of research."

Section 9 — The Future of InsightSwarm

Planned features:

- Multilingual support (Hindi, Marathi, Spanish...)
 - Image and video fact-checking
 - Browser extension
 - Social media warning-label API
 - Mobile app
 - WhatsApp bot integration
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Closing Note

InsightSwarm was born out of frustration — frustration that misinformation spreads faster than truth, that existing tools are either too slow or too unreliable, and that ordinary people have no easy way to check what they read.

The core idea is simple: **Do not trust one AI's opinion. Make two AIs debate. Check every source they use against the real web. Then let a judge decide.**

It sounds complicated, but it runs in under a minute and costs nothing. That is the whole point.

We hope this guide helped you understand the project. For full technical docs, see the [README.md](#) in the project repository.

END OF GUIDE

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